

From Hidden Order to Identifiable Physics: A Target-First Observability Calculus with Admissible Ambiguity and Dual Obstructions

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Experimental design is often organized around an available instrument and only afterward around the claim the resulting data are meant to support. We formulate the inverse workflow. A target quantity is specified first; the source state, physical response, probe projection, ensemble averaging, instrument transfer, and nuisance structure are then written as a composed forward model; and the experiment is designed backward to remove observational equivalences that can change the target. Target-oriented design itself is established in c -optimal design and goal-oriented optimal experimental design; the present contribution is a physics-specific observability workflow that places an explicit source-state gate, physical factorization, dual counterexample construction, forbidden reversal sectors, and common-state closure around that target. For an exact linear experiment with stacked map A_E and linear target C , we define the residual structural ambiguity as $\mathcal{O}_C(E) = C(\ker A_E)$ and its dimension d_C . Because d_C is binary for a scalar target, we additionally define the data- and physics-conditioned admissible target set $\mathcal{O}_C(E; y, \Omega)$ and its diameter, yielding a continuous ambiguity contraction before exact closure. A worked counterexample shows that a high-amplitude measurement can have zero target design gain while a much smaller signal exactly closes the target. For nonlinear noisy experiments we pair structural closure with target-conditioned practical margins derived from nuisance-projected sensitivities. Controlled reversals create both positive witness channels and forbidden sectors for obstruction tests. Cross-probe closure additionally requires a common latent state. The resulting protocol proceeds from target to state gate, forward map, admissible ambiguity, dual counterexamples, reversal channels, complementary probes, practical margin, and attempted falsification.

I. INTRODUCTION

A detector records data; a paper makes a physical claim. The gap between those two objects is where many experimental ambiguities live. A nonzero signal can be reproducible yet fail to identify the quantity assigned to it, while a null channel can coexist with a nonzero physical response that has been projected out, domain averaged, or transferred below practical resolution. The design problem is therefore not simply to maximize signal. It is to choose measurements that distinguish states whenever those states imply different values of the intended target.

The mathematical pieces of this statement are classical. Observability identifies state directions that cannot be reconstructed from outputs; structural identifiability asks whether different parameter values generate the same observations; inverse-problem theory separates nonuniqueness from instability and regularization; c -optimal design explicitly targets the variance of a chosen linear functional; goal-oriented optimal experimental design minimizes uncertainty in a specified quantity of interest; and Fisher information and model-manifold geometry diagnose locally weak parameter combinations.[1–11, 13, 14]

The difficulty in physics is usually not absence of these tools but fragmentation across layers. Source physics, response formation, probe symmetry, domain population, resolution, calibration, and inference are often discussed in different languages. A useful experimental workflow

must preserve those distinctions while still answering a simple practical question: *what should be measured next if the goal is to establish or falsify a specific claim?*

A companion measurement-map analysis develops the forward geometry of observational equivalence, cumulative null spaces, dual witness spaces, common-state probe closure, and practical near-nulls. The present paper deliberately does not re-derive that geometry in full. Instead, it turns the geometry around. The experimentalist starts from a target and designs backward toward a set of measurements that destroys the target-relevant ambiguity while simultaneously constructing channels capable of falsifying the attribution.

We organize this backward design around the rule

$$\begin{array}{c} \text{target} \rightarrow \text{state gate} \rightarrow \text{map} \\ \rightarrow \text{obstructions} \rightarrow \text{witnesses} \rightarrow \text{falsification} \end{array} \quad (1)$$

The result is a portable observability calculus for experiment design rather than another material-specific mechanism.

II. THE TARGET-FIRST DESIGN PROBLEM

Let $x \in X$ denote a latent physical state or parameter vector, and let the target claim be represented by a map

$$f : X \rightarrow Z. \quad (2)$$

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For a linearized problem we write $f(x) = Cx$. A candidate experiment e supplies a measurement map

$$\mathcal{M}_e = \mathcal{T}_e \circ \mathcal{A}_e \circ \mathcal{P}_e \circ \mathcal{R}_e \circ \mathcal{S}_e, \quad (3)$$

where $\mathcal{S}$ denotes source/state preparation, $\mathcal{R}$ response formation, $\mathcal{P}$ probe projection, $\mathcal{A}$ ensemble or domain averaging, and $\mathcal{T}$ resolution and detector transfer. Nuisance variables may enter any stage.

The target is identifiable from experiment e precisely when it is constant on the fibers of $\mathcal{M}_e$. In a finite-dimensional linear model with measurement matrix A_e , this becomes

$$\ker A_e \subseteq \ker C. \quad (4)$$

Equation (4) is intentionally target specific. An experiment need not reconstruct every microscopic coordinate. It need only distinguish states that would change the claim.

III. STRUCTURAL DESIGN GAIN: REMOVE THE RIGHT NULLS

Suppose a set E of exact common-state experiments has stacked linear map

$$A_E = \begin{bmatrix} A_{e_1} \\ A_{e_2} \\ \vdots \end{bmatrix}. \quad (5)$$

The residual target ambiguity is the image of the remaining null space under the target:

$$\mathcal{O}_C(E) = C(\ker A_E). \quad (6)$$

The target is structurally closed exactly when $\mathcal{O}_C(E) = \{0\}$.

For finite-dimensional linear targets define

$$d_C(E) = \dim \mathcal{O}_C(E) = \dim C(\ker A_E). \quad (7)$$

As in the underlying measurement geometry, d_C is not a probability, confidence level, or evidence score. It counts exact target dimensions that remain unresolved in the stated model.

Now let B be a candidate additional measurement. Its *structural design gain* relative to the current set is

$$\Delta_C(B | E) = d_C(E) - d_C(E \cup B) \geq 0, \quad (8)$$

provided all measurements refer to a common latent state. A candidate probe with $\Delta_C = 0$ may be scientifically useful for calibration or model checking, but it does not improve exact identification of the stated target. Conversely, a probe can be highly valuable even if it leaves many unrelated coordinates unconstrained, provided it destroys the null directions that change C .

The same criterion exposes redundancy. Two nominally different instruments can probe nearly the same witness subspace. Repeated agreement between them may increase precision while leaving d_C unchanged. Experimental closure should therefore be discussed in terms of residual ambiguity, not the number of positive techniques.

IV. EXACT CLOSURE VERSUS ADMISSIBLE-SET CONTRACTION

The dimension d_C is deliberately coarse. For a scalar target $C : X \rightarrow \mathbb{R}$, any nonzero linear image $C(\ker A_E)$ has dimension one. Therefore an experiment that shrinks a physically admissible target interval from $[-100, 100]$ to $[-10^{-2}, 10^{-2}]$ can leave $d_C = 1$ even though it has reduced the scientifically relevant ambiguity by four orders of magnitude. Exact structural closure and quantitative contraction must therefore be reported separately.

Let $\Omega \subseteq X$ be a declared physically admissible state set and let y_E denote the observed data (or an accepted data-consistency set). Define

$$\mathcal{O}_C(E; y_E, \Omega) = \{Cx : x \in \Omega, A_E x = y_E\}. \quad (9)$$

For a scalar target, a natural ambiguity measure is its width

$$w_C(E; y_E, \Omega) = \text{diam } \mathcal{O}_C(E; y_E, \Omega). \quad (10)$$

For vector targets one may use a normed diameter, volume, or a decision-relevant loss functional; the metric and physical scaling must be declared because these quantities are not coordinate free.

For a candidate probe B with observed result y_B , define the admissible-set contraction

$$\Delta_C^{\text{diam}}(B | E) = w_C(E; y_E, \Omega) - w_C(E \cup B; y_E, y_B, \Omega). \quad (11)$$

Adding consistent information can only shrink the feasible state set, so the target ambiguity set is nested and $\Delta_C^{\text{diam}} \geq 0$. This quantity does not replace d_C : the former records bounded progress before uniqueness, while the latter records exact structural closure.

A decision-theoretic version is also possible. For a loss $L(a, z)$ on target values z , define the worst-case residual risk

$$R_C(E) = \inf_a \sup_{z \in \mathcal{O}_C(E; y_E, \Omega)} L(a, z). \quad (12)$$

This makes explicit that a design can be scientifically useful before exact identification if it contracts the ambiguity enough for the decision at hand.

V. THE STATE GATE COMES BEFORE DOWNSTREAM CLOSURE

The kernel-intersection rule is valid only for measurements of a common latent state. Let measurement i

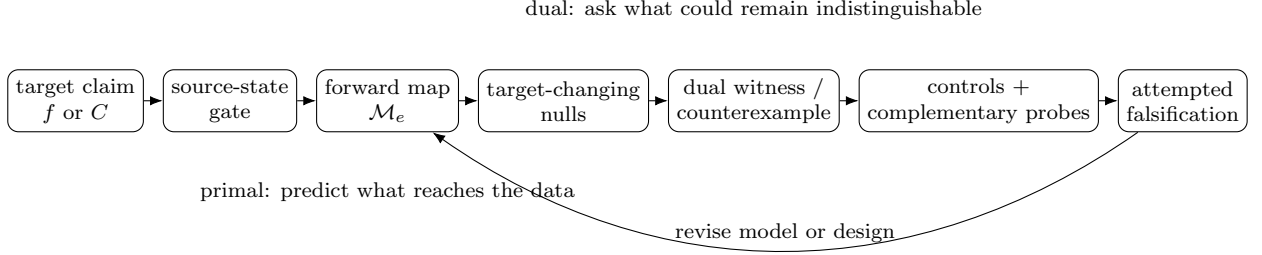


FIG. 1. Target-first experimental design. The workflow starts from the claim rather than from a preferred instrument. The forward model predicts the data, while the dual path constructs target-changing states that the current experiment cannot distinguish and then designs measurements that remove those ambiguities. The process ends with attempted falsification and loops back when the source model or measurement design fails. Schematic TikZ code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the author specified the scientific content and verified the final rendering.

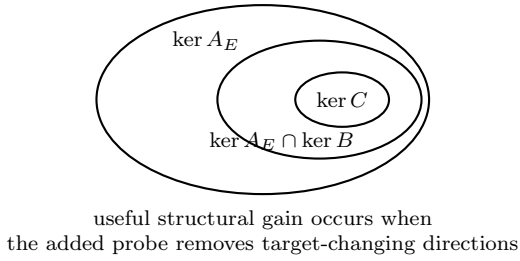


FIG. 2. Structural design gain. A complementary probe is useful for a target when the residual common-state null intersection moves toward $\ker C$. Full microscopic reconstruction is unnecessary; closure is reached once every remaining indistinguishable displacement is target preserving. Schematic TikZ code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the author specified and verified the content.

depend on additional preparation or material variables ζ_i :

$$y_i = \mathcal{M}_i(x, \zeta_i). \quad (13)$$

If domain population, strain, stoichiometry, temperature, surface reconstruction, history, or phase differs across measurements, one cannot silently replace Eq. (13) by $y_i = \mathcal{M}_i(x)$ and intersect the corresponding null spaces. Either the experiment must establish common-state conditions or the ζ_i must be promoted into the latent model.

This requirement gives a directional rule for evidence. A downstream response cannot repair a contradicted upstream source-state attribution. If the target claim logically requires a prerequisite state H_S , then the design has two distinct obligations:

H_S must survive an independent state gate
before downstream response closure is claimed

(14)

The state gate is not a demand that every microscopic detail be known. It is a demand that the material or preparation state assumed by the downstream forward

model be independently compatible with the measurements being combined.

This distinction is particularly important in cross-material or thin-film/bulk comparisons, where superficially similar signatures may arise from physically different latent states. A state-gated workflow prevents downstream observables from being used as votes that numerically outweigh a failed prerequisite.

VI. DUAL OBSTRUCTIONS: DESIGN THE EXPERIMENT THAT COULD PROVE YOU WRONG

Target identifiability has a constructive dual. For a current stacked map A_E , any vector

$$v \in \ker A_E, \quad Cv \neq 0, \quad (15)$$

is an explicit counterexample to the claim that the existing data identify C . The most useful next experiment is often the one that is maximally sensitive to such a counterexample.

The backward-design procedure is therefore:

1. find a target-changing null direction or measurement fiber;
2. identify the physical transformation represented by that ambiguity;
3. choose a control, geometry, or complementary probe that couples differently to the two states;
4. include a negative or forbidden channel that should remain null if the attribution is correct.

This dual requirement changes the role of controls. A negative control is not a secondary quality-assurance measurement. It is part of the identifying structure. A model that predicts a target contribution in one symmetry sector should often predict its absence in another; measuring both sectors tests the assignment rather than merely the existence of a signal.

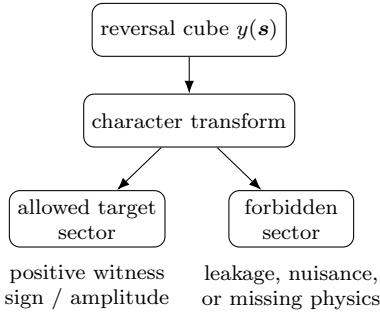


FIG. 3. Reversal-space coding produces a target witness and a dual obstruction simultaneously. The desired contribution occupies an allowed character coefficient; sectors forbidden by the model test leakage, uncontrolled state changes, or missing physics. Schematic TikZ code was developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the author specified and verified the scientific content.

VII. REVERSAL SPACE AS AN EXPERIMENTAL CODING BASIS

Controlled transformations provide an especially clean implementation of dual design. Any complete N -factor binary acquisition cube admits an exact Walsh–Hadamard factorial decomposition. Interpreting those factors as physical $(\mathbb{Z}_2)^N$ characters, however, additionally requires the implemented reversals to behave as involutions and to commute on the relevant prepared state; otherwise the coefficients remain valid factorial contrasts but not group characters. The response is decomposed as

$$\hat{y}_A = \frac{1}{2^N} \sum_{\mathbf{s}} \left(\prod_{i \in A} s_i \right) y(\mathbf{s}). \quad (16)$$

Contributions with different transformation signatures occupy orthogonal coefficients. The desired target can therefore be placed in a parity sector separated from dominant even backgrounds, while coefficients forbidden by the model become obstruction channels.

The limitation is equally informative. If two microscopic mechanisms transform identically under every available control, reversal algebra alone cannot separate them. A new independent control or probe is then required. Failure to distinguish those mechanisms is an identifiability result, not a reason to assign one by preference.

VIII. PRACTICAL DESIGN AFTER STRUCTURAL VIABILITY

Structural identifiability is necessary but does not guarantee an estimable experiment. Let $m(\theta)$ denote model predictions, $J = \partial m / \partial \theta$ the local sensitivity, Σ the measurement covariance, and B nuisance sensitivities. Whitening and projecting away the nuisance tangent space

gives

$$J_{\text{eff}} = (I - Q_B) \Sigma^{-1/2} J, \quad (17)$$

where Q_B projects onto the whitened nuisance span.

For a linearized target $C \delta \theta$, define the *target practical margin*

$$\mu_C(J_{\text{eff}}) = \inf_{\|Cv\|_2=1} \|J_{\text{eff}} v\|_2. \quad (18)$$

If $\mu_C = 0$, a target-changing local direction is unobserved after nuisance projection. A small nonzero value indicates a practical near-null: the target is structurally present but weakly distinguished relative to the adopted covariance and model. The numerical magnitude of μ_C is experiment dependent and has no universal threshold.

Equation (18) suggests a two-stage design discipline. First expose exact target-changing nulls using d_C or an equivalent structural criterion and, when exact closure is not yet possible, report admissible-set contraction using Eq. (11). Then optimize practical precision, conditioning, acquisition cost, dose, time, or other utilities. This second stage is where classical c -optimal design and Bayesian or goal-oriented optimal design naturally enter: they optimize uncertainty in a specified functional or quantity of interest once the model, admissible state, and design class have been declared.[7–11] The present calculus is therefore not a replacement for target-oriented OED; it places physics-specific state gates, dual counterexamples, reversal obstructions, and common-state closure around it.

Regularization and priors belong to a third category. They can stabilize an estimate or choose among likelihood-equivalent states, but they do not create a witness direction absent from the data. When a unique estimate depends on a prior while the likelihood remains flat along a target-changing direction, the correct statement is prior-conditional inference rather than data identification.[4, 5, 15]

IX. WORKED COUNTEREXAMPLE: MAXIMUM SIGNAL CAN BE THE WRONG EXPERIMENT

Consider a latent state $x = (\theta, \nu)^T$ with scalar target $C = [1 \ 0]$, so the claim concerns θ while ν is a nuisance coordinate. Suppose the existing experiment measures the nuisance exactly,

$$A_E = [0 \ 1]. \quad (19)$$

The current null space is $\ker A_E = \text{span}\{(1, 0)^T\}$, hence $d_C(E) = 1$.

Compare two candidate measurements. A high-amplitude probe responds only to the nuisance,

$$B_{\text{sig}} = [0 \ 100], \quad (20)$$

whereas a much smaller target-sensitive probe measures

$$B_{\text{tar}} = [1 \ 0.01]. \quad (21)$$

For the representative state $x = (0.01, 1)^T$, the raw responses are

$$|B_{\text{sig}}x| = 100, \quad |B_{\text{tar}}x| = 0.02. \quad (22)$$

A signal-maximizing rule therefore prefers B_{sig} by a factor of 5000 in amplitude. Yet

$$\ker \begin{bmatrix} A_E \\ B_{\text{sig}} \end{bmatrix} = \text{span}\{(1, 0)^T\}, \quad (23)$$

so $\Delta_C(B_{\text{sig}} | E) = 0$. By contrast,

$$\text{rank} \begin{bmatrix} A_E \\ B_{\text{tar}} \end{bmatrix} = 2, \quad (24)$$

and the target is exactly identified: $\Delta_C(B_{\text{tar}} | E) = 1$. The low-amplitude experiment is therefore the structurally decisive measurement.

This example does not argue against signal-to-noise optimization. It shows the required ordering: a large response in an already observed nuisance direction cannot substitute for a new target witness. Once a design has positive target gain, conventional variance or utility optimization should choose among the viable designs.

X. A THIRTEEN-STEP OBSERVABILITY PROTOCOL

The complete workflow is summarized below. It is intentionally ordered so that later precision cannot compensate for an earlier logical failure.

1. **State the target.** Specify the scalar, sign, tensor component, rank, parameter combination, or hypothesis to be established.
2. **State the prerequisite state.** Identify the material, domain, phase, preparation, or symmetry assumptions required by the claim.
3. **Build the state gate.** Choose at least one measurement or control capable of falsifying that prerequisite rather than inferring it only from the downstream target channel.
4. **Factor the forward map.** Separate source, response, probe projection, ensemble/domain averaging, and instrument transfer as far as the physics allows.
5. **Include nuisance and state variables.** Calibration, leakage, backgrounds, preparation variables, and model-discrepancy terms belong in the map.
6. **Find target-changing fibers or nulls.** Determine whether two admissible states can produce the same current data while changing the target.

7. **Quantify admissible ambiguity.** For bounded physical state sets, report target width, diameter, volume, or decision loss in addition to exact structural dimension.

8. **Construct a dual counterexample.** Make at least one such ambiguity explicit or identify a symmetry sector predicted to vanish.

9. **Encode controlled reversals.** When possible, place the target in a transformation sector separated from dominant backgrounds and record forbidden sectors as diagnostics.

10. **Add complementary probes selectively.** Prefer measurements with $\Delta_C > 0$ or their nonlinear analogue; do not confuse repeated witnesses with new closure.

11. **Verify common-state compatibility.** Combine probes only after showing that they refer to one latent state or after enlarging the model to include the differences.

12. **Quantify practical margin.** Report target-relevant singular values, profile widths, Fisher information, resolution sensitivity, or equivalent conditioning diagnostics.

13. **Attempt falsification.** Test the state gate, forbidden reversal sectors, competing mechanisms, and source-model alternatives. Preserve the negative result if the claim fails.

The protocol is compatible with, and deliberately downstream of, classical target-oriented design. c -optimal design already optimizes a chosen linear functional, and goal-oriented OED already targets a quantity of interest rather than the full parameter vector.[7, 11] The additional question posed here is physics-specific: *which target-changing ambiguity survives the declared source state, probe map, reversals, nuisance structure, and common-state assumptions?* Exact closure and admissible-set contraction are assessed before precision, time, cost, or information utility are optimized.

XI. WORKED PATTERN: ONE MATERIAL, THREE DIFFERENT TARGETS

Altermagnetic MnF_2 provides a useful example because several experiments interrogate related physics while measuring different mathematical objects.[19–23]

Consider three targets:

$$C_1 : \text{signed weak exchange coordinate}, \quad (25)$$

$$C_2 : \text{dynamical antisymmetric response}, \quad (26)$$

$$C_3 : \text{static multipole content or rank}. \quad (27)$$

For C_1 , the Hamiltonian and bond-label convention must be stated explicitly. They are related by a physical model

TABLE I. Target-first experimental-design checklist. The rightmost column is the dual obligation paired with each forward design step.

Design object	Forward question	Dual / falsification question
Target	What exact quantity is the paper claiming?	What pair of states would change the claim while preserving present data?
Admissible ambiguity	How large is the remaining target set before exact closure?	Did the next probe contract the physical target interval even if d_C is unchanged?
State gate	What prerequisite state does the forward model assume?	Which independent measurement could show that state is absent or different?
Probe map	Which response component reaches the detector?	What target component lies in the probe null?
Averaging	Which domains or orientations are combined?	Can opposite populations cancel the target sector?
Transfer	Which spectral or spatial directions survive resolution and binning?	Which target contrast is weak or erased by the transfer operator?
Reversals	Which character sector should contain the target?	Which character coefficients should remain forbidden?
Complementary probe	Which residual null direction does the new probe remove?	Does it actually reduce d_C , or only repeat an existing witness?
Practical margin	How far is the target from a near-null direction?	Does the claim depend on a prior, regularizer, or calibration choice?
Closure	Do all remaining ambiguities preserve the target?	Can a competing mechanism still reproduce the complete allowed/forbidden pattern?

but are not the same functional of the latent state. A field/partner-momentum protocol can increase local sensitivity to the signed weak exchange coordinate; polarized inelastic neutron scattering can isolate a dynamical chiral channel; polarized diffraction can reconstruct static magnetization density and its multipolar components. None of these measurements is automatically a substitute for the others.

The target-first calculus therefore forbids three shortcuts. First, a visible magnon splitting does not by itself identify the signed microscopic exchange that contributes to it. Second, a polarized inelastic chiral signal does not by symmetry alone determine a static multipole rank. Third, a static multipolar reconstruction does not by itself prove the detailed dynamical response observed by another probe. Closure requires a model in which the corresponding maps are jointly evaluated and in which the measurements refer to compatible material/domain states.

Controlled reversals then provide the dual checks. Reversing polarization, partner momentum, magnetic domain, or field can produce predicted sign patterns; sectors expected to vanish act as systematics tests. The point is not that MnF_2 supplies a universal mechanism. It supplies a compact example in which source coordinate, response tensor, probe projection, domain state, and instrumental transfer can be kept distinct while still being tested in one physical system.

The current literature also illustrates why publication status and evidence layer should not be conflated. The polarized-neutron MnF_2 work of Faure *et al.* remains publicly listed as “to appear in Science” at the September 12, 2026 cutoff, while the real-space multipole reconstruction of Kibalin *et al.* and the field-readout proposal of

Jiang *et al.* remain recent preprints.[21–23] The calculus is independent of those publication-status details; the claims made from each source should remain limited to the observable and model actually established.

XII. IMPLEMENTATION AND REPORTING STANDARD

A practical software implementation need not solve a symbolic global inverse problem in every case. The workflow can be implemented at increasing levels of sophistication:

1. exact linear kernels and row spaces when the model is linear;
2. Jacobians, nuisance projection, and singular vectors for local nonlinear design;
3. profile likelihood or posterior predictive analysis for nonlinear practical identifiability;[13, 15]
4. symbolic or differential-algebraic identifiability methods for structured dynamical models;[16]
5. finite-group character transforms for controlled reversal experiments;[17]
6. direct model-discrimination experiments when no common latent state can fit the observations.

For reproducibility, a paper using the calculus should report at minimum: the target; the assumed source/material state; the factorized forward map; the nuisance variables; one explicit target-changing ambiguity

or evidence that none remains; the reversal or complementary probe that addresses it; a practical conditioning diagnostic; and at least one attempted falsification. This reporting standard makes negative results scientifically useful because it identifies where the inference failed rather than collapsing every failure into “no signal.”

XIII. RELATION TO ESTABLISHED THEORY AND NOVELTY BOUNDARY

No individual mathematical ingredient introduced here is claimed as new. Unobservable subspaces, quotienting by observational equivalence, structural identifiability, annihilator/row-space duality, Fisher-information rank, finite-group projectors, Walsh–Hadamard analysis, c -optimal design, and goal-oriented optimal experimental design are established.[1, 3, 6, 7, 10–12, 17] In particular, the idea of designing measurements around a specified linear functional or quantity of interest predates the present work.

The contribution claimed here is narrower: a physics-facing synthesis that composes target-oriented design with (i) an explicit material/source-state gate; (ii) a physically factorized forward map; (iii) exact and admissible-set target ambiguity reported separately; (iv) constructive dual counterexamples and forbidden channels; (v) path-validated reversal sectors as experimental coding; and (vi) common-state requirements for cross-probe closure. The worked signal-maximization counterexample demonstrates why this bookkeeping can select a different experiment from a response-amplitude criterion even in a two-parameter linear model.

The framework does not prove that a chosen physical factorization is unique, that a local Jacobian establishes global identifiability, that the declared admissible set is complete, or that a nonzero structural design gain implies an experimentally feasible measurement. Closure is always conditional on the forward-model class and state constraints being tested. Nor does the framework supply a universal scalar score by which unrelated experiments should be ranked. Structural closure, admissible-set contraction, practical conditioning, cost, safety, and scientific utility are different axes and should remain visible rather than being collapsed into one number.

XIV. DISCUSSION

The central design shift is from signal-centered to claim-centered experimentation. Signal optimization asks where a detector responds strongly. Claim-centered design asks which measurement distinctions are necessary for the paper’s intended inference. Those objectives often overlap, but they are not identical.

This view changes how complementary probes are justified. A second technique is not valuable merely because it produces another positive result. It is valuable when

it adds a witness direction absent from the first technique, removes a target-changing ambiguity, calibrates a nuisance direction, or tests a source-state prerequisite. Likewise, a negative control is not a ceremonial check; it is often the dual half of the identification argument.

The framework also clarifies where an experimental program should stop. If a source-state gate fails, downstream signatures should not be used to rescue the source hypothesis. If a target remains structurally unidentifiable after every feasible probe, the correct result is an equivalence class rather than an arbitrarily selected microscopic explanation. If a unique estimate is created only by a prior or regularizer, that dependence should be reported explicitly. And if independent datasets cannot be reproduced by any common latent state, the failure belongs to the model or state hypothesis rather than to another category of detector null.

The desired endpoint is therefore not maximal certainty. It is a claim whose surviving ambiguity has been stated, whose witness channels have been designed deliberately, and whose most plausible alternatives have been given a fair opportunity to fail the claim.

XV. CONCLUSION

A portable experimental-design calculus follows from one principle: design backward from the target while modeling forward from the physics. For exact common-state linear measurements, the residual target ambiguity is

$$\mathcal{O}_C(E) = C(\ker A_E), \quad (28)$$

and closure occurs when this set vanishes. For bounded physical state sets, the companion quantity $\mathcal{O}_C(E; y, \Omega)$ records continuous contraction of the still-admissible target values before exact uniqueness. A new probe is structurally decisive when it removes a target-changing null and can still be quantitatively valuable when it contracts the admissible target set; a dual counterexample v with $A_E v = 0$ and $Cv \neq 0$ shows exactly why the current design is insufficient. Controlled reversals convert symmetry into experimental coding, creating allowed target channels and forbidden obstruction channels. Nonlinear noisy experiments replace exact kernels by nuisance-projected local sensitivities and target-conditioned practical margins, while common-state checks prevent spurious closure across physically different samples or preparation states.

The capstone workflow is therefore

$$\begin{aligned} & \text{target} \rightarrow \text{state gate} \rightarrow \text{forward map} \\ & \rightarrow \text{target nulls / admissible set} \rightarrow \text{dual obstruction} \\ & \rightarrow \text{reversals / complementary probes} \\ & \rightarrow \text{practical margin} \rightarrow \text{attempted falsification} \end{aligned}$$

(29)

The purpose is not to guarantee a preferred interpretation. It is to make explicit what the experiment can identify,

what it cannot, and which next measurement would most directly decide between the surviving alternatives.

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cited literature and independent algebraic reasoning, inspected the rendered manuscript and figures, and accepts full responsibility for all claims, equations, references, figures, and conclusions.

DATA AVAILABILITY

No new experimental data were generated for this theoretical and methodological work.

COMPETING INTERESTS

The author declares no competing interests.

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